



BHARATI VIDYAPEETH'S

INSTITUTE OF COMPUTER APPLICATIONS & MANAGEMENT

(Affiliated to Guru Gobind Singh Indraprastha University, Approved by AICTE, New Delhi)

Object Oriented Software Engineering Lab

(MCA-164)

Practical File

Submitted To:

Dr. Ritika Wason

(Deputy Director & Associate Professor)

BVICAM

Submitted By:

Himneesh Kalra (01811604425)

MCA 2nd Sem

Sec A

INDEX

S.No	Problem Description	Date of Execution	Sign
Q1.	Create a Use Case Diagram for a Digital Sound Recorder with the following main features: 1. The recorder stores up to 10 messages 2. Each message is max. 2 minutes long 3. The user can record message 4. Recording of a message end after 2 minutes or when the user stops recording 5. Recording destroys the original message at chosen slot 6. Sufficient level of battery is checked before recording message 7. Message of a given slot can be replayed 8. Sufficient level of battery is checked before replaying message 9. Messages can be locked/unlocked 10. Locked messages cannot be deleted or over-written by recording to the same slot User uses LCD display and buttons to interact with recorder.		
Q2.	Formulate use case descriptions for 5 major use cases identified in the digital sound recorder.		
Q3.	As the head of Information Technology at Acme Airlines, you are tasked with building a new booking system to replace the existing system. Acme needs a new system to allow employees to book ticket electronically and check the status of the booked tickets. The new system will be state of the art and will have a Windows-based desktop interface to allow employees to enter booking details and view employee details. The system will run on individual employee desktops throughout the entire company. For reasons of security and auditing, employees can access only their data. The employees would only be able to view their information, and for any changes they would have to send an E-Mail to the administrator. Administrator would only have the right to make any changes in the records. For the cost being a factor Acme wants to use their old server with the existing database. The system will retain information on all employees in the company. The administrator maintains employee information. He is responsible for adding new employees, deleting employees and changing all employee information such as name, address, and paycheck generation, as well as running administrative reports.		
Q4.	Consider the following use-case of a travel agency. Use-Case Name: Ticket Purchasing Description:		

	1. The use-case begins when the customer calls the travel agency to ask it to issue a ticket that (s)he has booked. 2. The travel agency operator asks the customer to give his/her booking number. 3. The customer gives the booking number. 4. The operator types in the booking number and the flight reservation system displays the details of the reservation made. 5. The operator asks the customer to confirm the details of the reservation made. 6. The customer confirms the reservation made. 7. The operator asks the customer for a credit card number. 8. The customer gives his/her credit card number. 9. The operator types in the customer's credit card number and when the system confirms that the credit card transaction has been authorized (s)he asks the system to print the tickets, the details of the flights, and an invoice. 10. When the system confirms that the requested items have been printed, the operator informs the customer that the tickets have been issued. Think of at least two alternative courses of events for this use-case. Describe the alternative courses as separate use-cases. Modify the description of the original usecase to make evident where exactly these alternative use-cases may be called and under which conditions. Create a use-case diagram to illustrate the relationships between the alternative use-cases and the use-case given.		
Q5.	Create a Class Diagram for a Digital Sound Recorder with the following main features: 1. The recorder stores up to 10 messages 2. Each message is max. 2 minutes long 3. The user can record message 4. Recording of a message end after 2 minutes or when the user stops recording 5. Recording destroys the original message at chosen slot 6. Sufficient level of battery is checked before recording message 7. Message of a given slot can be replayed 8. Sufficient level of battery is checked before replaying message 9. Messages can be locked/unlocked 10. Locked messages cannot be deleted or over-written by recording to the same slot. User uses LCD display and buttons to interact with recorder.		

Q6.	Forward engineer the model obtained in Q5 to automatically generate equivalent Java stub Code?		
Q7.	Purpose: Describe major services (functionality) provided by a hospital's reception. Summary: Hospital Management System is a large system including several subsystems or modules providing variety of functions. Hospital Reception subsystem or module supports some of the many job duties of hospital receptionist. Receptionist schedules patient's appointments and admission to the hospital, collects information from patient upon patient's arrival and/or by phone. For the patient that will stay in the hospital ("inpatient") she or he should have a bed allotted in a ward. Receptionist may also receive and process patient's payments, record them and provide receipts. She also is responsible to file insurance claims and generate medical reports. a. Make the use case diagram b. Provide description for at least 04 use cases c. Make the class diagram d. Generate Java stub code e. Create Object Diagram		
Q8.	Create an object diagram for a Digital Sound Recorder with the following main features: 1. The recorder stores up to 10 messages 2. Each message is max. 2 minutes long 3. The user can record message 4. Recording of a message end after 2 minutes or when the user stops recording 5. Recording destroys the original message at chosen slot 6. Sufficient level of battery is checked before recording message 7. Message of a given slot can be replayed 8. Sufficient level of battery is checked before replaying message 9. Messages can be locked/unlocked 10. Locked messages cannot be deleted or over-written by recording to the same slot. User uses LCD display and buttons to interact with recorder.		
Q9.	Create a package diagram for the Digital Sound Recorder in Q1.		
Q10.	Create an activity diagram for record message use case of Q1?		
Q11.	Create an activity diagram for any two use cases of Hospital Reception Subsystem.		
Q12.	Create a sequence diagram for the playback message use case of Q1?		

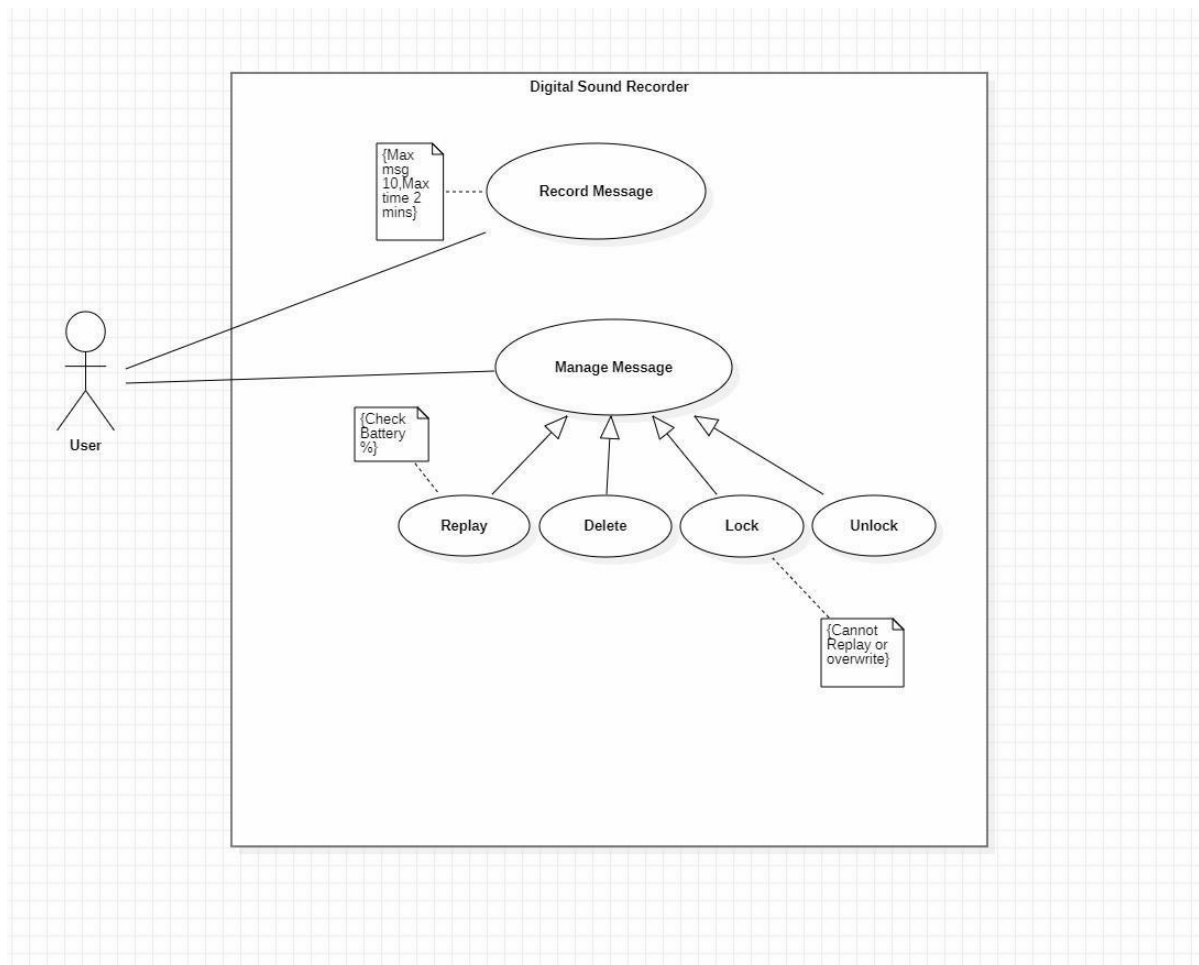
Q13.	Create a sequence diagram for any two use cases of Hospital Reception Subsystem.																																																		
Q14.	Create a state diagram for any two use cases of Hospital Reception Subsystem.																																																		
Q15.	Create a state diagram for a BANK ATM machine?																																																		
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	L	K	Cleanup	4		
	M	K	Send Back Bouncy Castle	1		
	N	M	Donate Unwanted Gifts	3		
	• Label hours 0 to 30 across the horizontal axis. • Mark a review stage at hour 14 to monitor the progress • Assume and illustrate that tasks A, B, C and D have been completed at hour 14 state which tasks are ahead and which tasks are behind schedule.					

Q1

Create a Use Case Diagram for a Digital Sound Recorder with the following main features:

1. The recorder stores up to 10 messages
2. Each message is max. 2 minutes long
3. The user can record message
4. Recording of a message end after 2 minutes or when the user stops recording
5. Recording destroys the original message at chosen slot
6. Sufficient level of battery is checked before recording message
7. Message of a given slot can be replayed
8. Sufficient level of battery is checked before replaying message
9. Messages can be locked/unlocked
10. Locked messages cannot be deleted or over-written by recorder



Q2 Formulate use case descriptions for 5 major use cases identified in the digital sound

Recorder

USE CASES	DESCRIPTION
Record Message	The user can record a message to a specific slot in the recorder, with a maximum duration of 2 minutes. The recording stops automatically after 2 minutes or when manually stopped by the user. Preconditions include having sufficient battery and the target slot being unlocked. Postconditions include storing the message in the selected slot and overwriting the previous message in the slot if it was unlocked.
Replay Message	The user can replay a stored message from a specified slot. Preconditions include having sufficient battery and the selected slot containing a message. Postconditions include playing the selected message through the recorder.
Lock/Unlock Message	The user can lock or unlock a message in a specified slot to prevent accidental deletion or overwriting. Preconditions require selecting a valid slot. Postconditions include marking the slot as locked or unlocked based on the user's action. Locked slots cannot be overwritten or deleted.
Check Battery Level	The system checks the battery level to ensure there is enough power to perform operations like recording or replaying a message. There are no specific preconditions as this is implicitly checked before recording or replaying. If the battery level is sufficient, the operation proceeds; otherwise, an error or warning is displayed.
Delete Message	The user can delete a message from a specific slot, but locked messages cannot be deleted. Preconditions require that the slot is not locked and contains a message. Postconditions include the removal of the message from the specified slot.

Q3

As the head of Information Technology at Acme Airlines, you are tasked with building a new booking system to replace the existing system. Acme needs a new system to allow employees to book ticket electronically and check the status of the booked tickets.

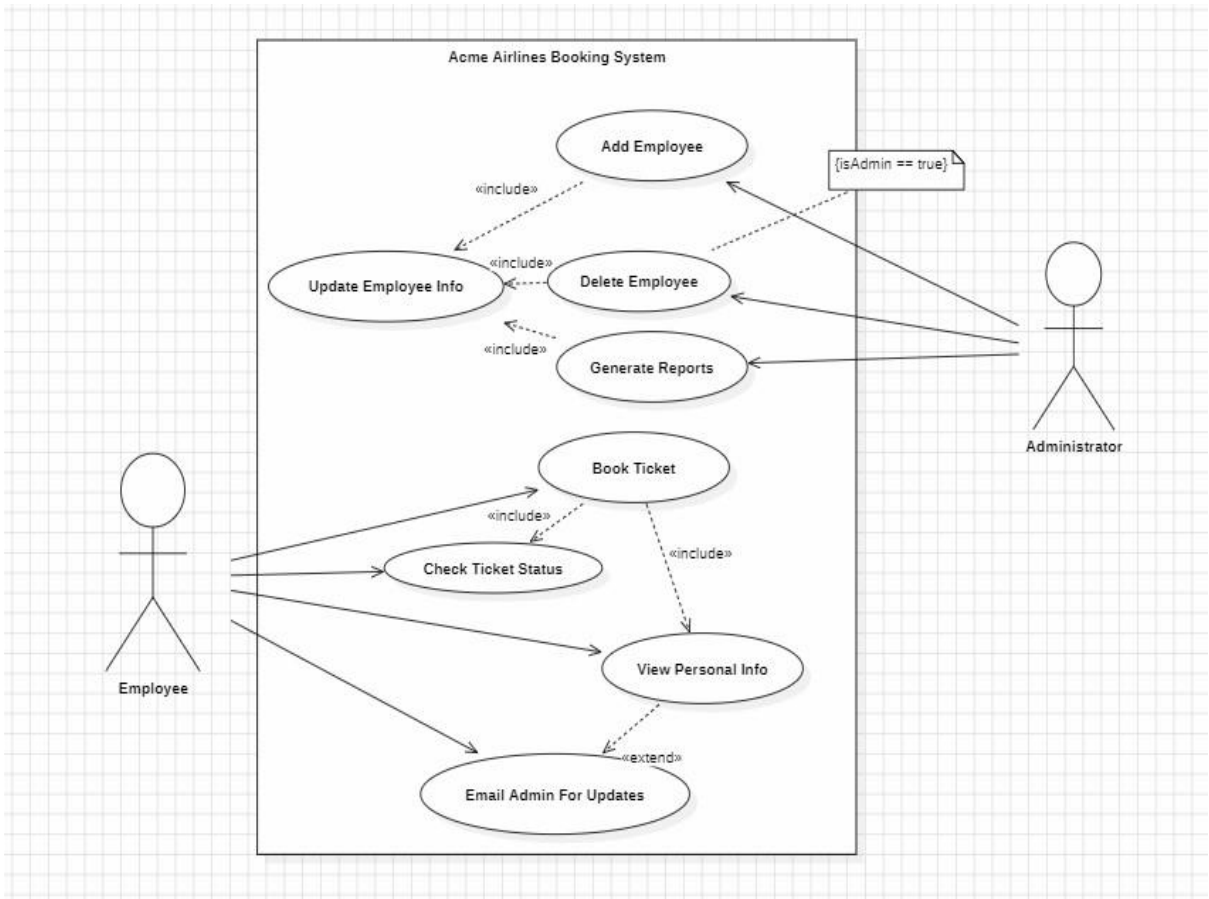
The new system will be state of the art and will have a Windows-based desktop interface to allow employees to enter booking details and view employee details. The system will run on individual employee desktops throughout the entire company.

For reasons of security and auditing, employees can access only their data. The employees would only be able to view their information, and for any changes they would have to send an

E-Mail to the administrator. Administrator would only have the right to make any changes in the records.

For the cost being a factor Acme wants to use their old server with the existing database. The system will retain information on all employees in the company.

The administrator maintains employee information. He is responsible for adding new employees, deleting employees and changing all employee information such as name, address, and paycheck generation, as well as running administrative reports.



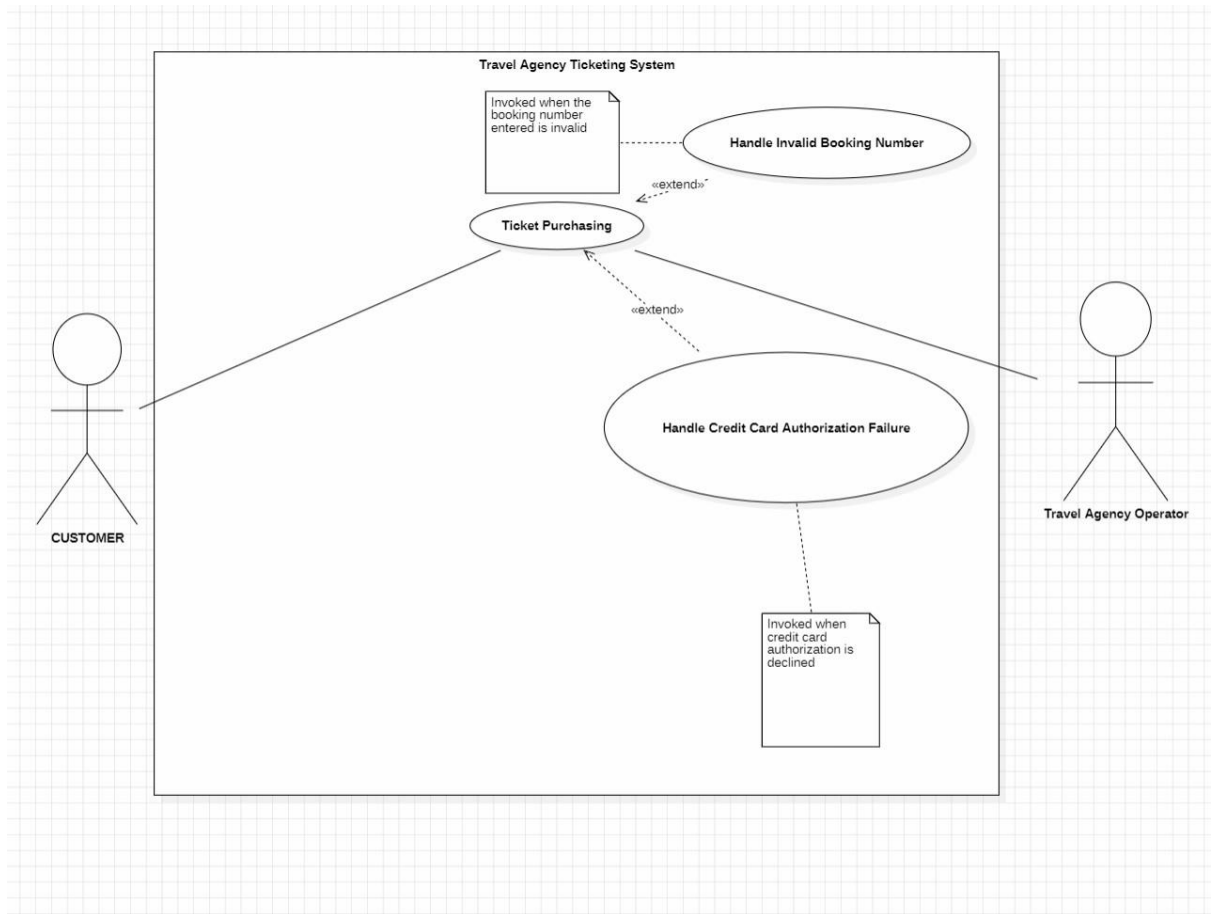
Q4 Consider the following use-case of a travel agency.

Use-Case Name: Ticket Purchasing

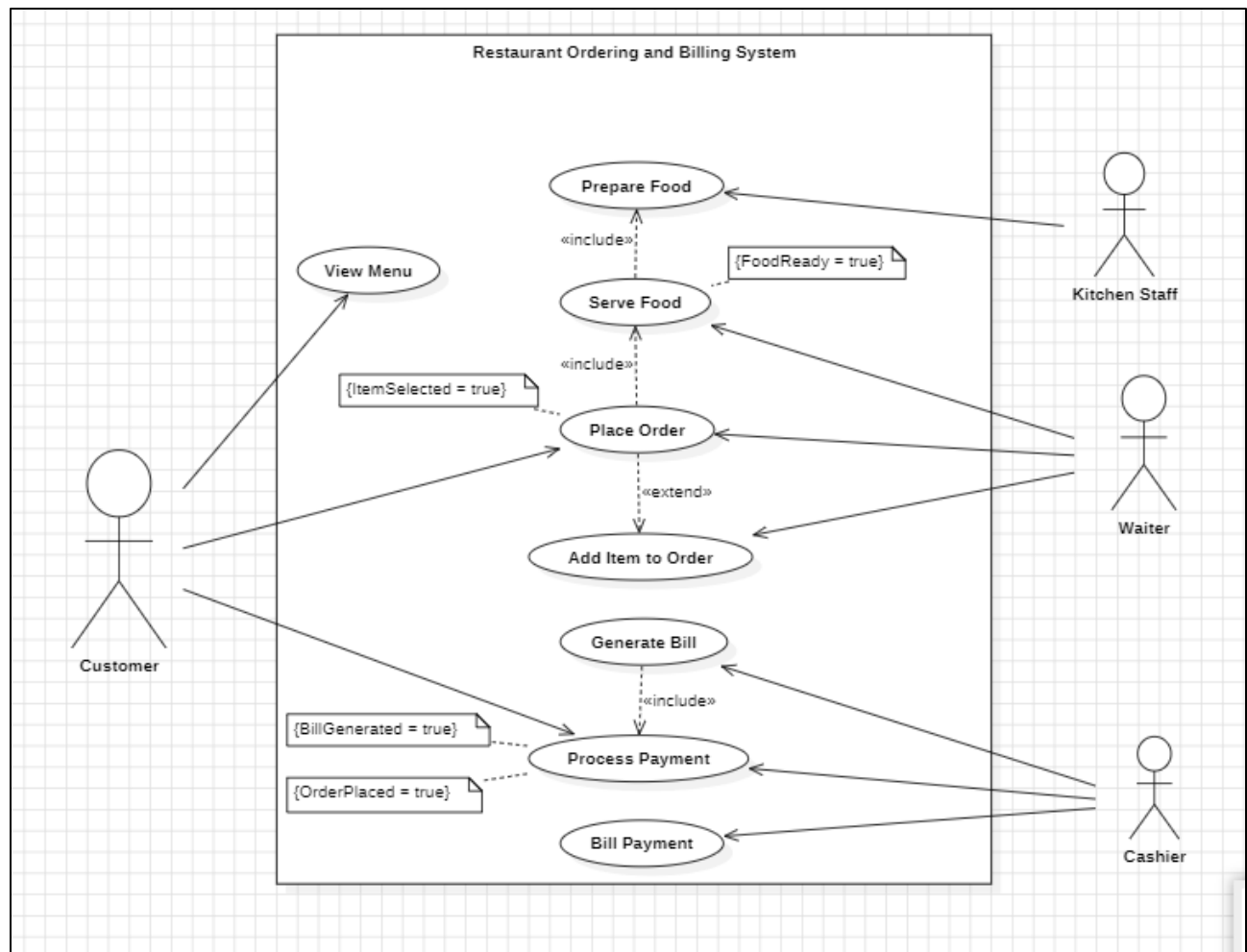
Description:

- 1. The use-case begins when the customer calls the travel agency to ask it to issue a ticket that (s)he has booked.**
- 2. The travel agency operator asks the customer to give his/her booking number.**
- 3. The customer gives the booking number.**
- 4. The operator types in the booking number and the flight reservation system displays the details of the reservation made.**
- 5. The operator asks the customer to confirm the details of the reservation made.**
- 5. The customer confirms the reservation made.**
- 6. The operator asks the customer for a credit card number.**
- 7. The customer gives his/her credit card number.**
- 8. The operator types in the customer's credit card number and when the system confirms that the credit card transaction has been authorized (s)he asks the system to print the tickets, the details of the flights, and an invoice.**
- 9. When the system confirms that the requested items have been printed, the operator informs the customer that the tickets have been issued.**

Think of at least two alternative courses of events for this use-case. Describe the alternative courses as separate use-cases. Modify the description of the original usecase to make evident where exactly these alternative use-cases may be called and under which conditions. Create a use-case diagram to illustrate the relationships between the alternative use-cases and the use-case given.



AQ1 Create a use case diagram for a restaurant ordering and billing system?

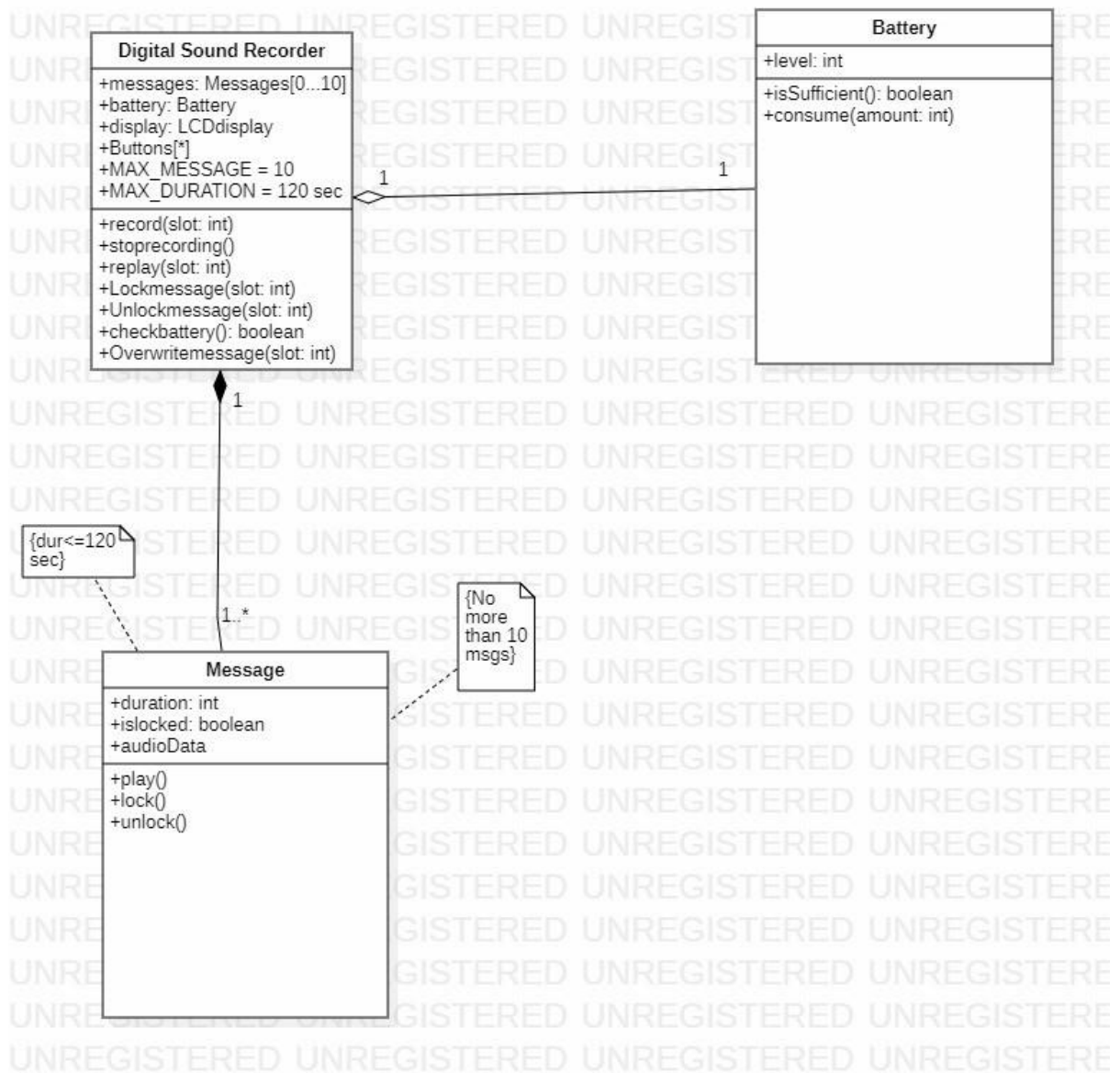


Q5

Create a Class Diagram for a Digital Sound Recorder with the following main features:

- 1. The recorder stores up to 10 messages**
- 2. Each message is max. 2 minutes long**
- 3. The user can record message**
- 4. Recording of a message end after 2 minutes or when the user stops recording**
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- 7. Message of a given slot can be replayed**
- 8. Sufficient level of battery is checked before replaying message**
- 9. Messages can be locked/unlocked**
- 10. Locked messages cannot be deleted or over-written by recording to the same slot**

User uses LCD display and buttons to interact with recorder



Q6 Forward engineer the model obtained in Q5 to automatically generate equivalent Java stub Code?

java file generator > Model				
Name	Date modified	Type	Size	
 Button	05-02-2025 14:48	Java Source File	1 KB	
 DigitalSoundRecorder	05-02-2025 14:48	Java Source File	2 KB	
 LCDDisplay	05-02-2025 14:48	Java Source File	1 KB	
 Message	05-02-2025 14:48	Java Source File	2 KB	

```
import java.io.*;
import java.util.*;

/**
 *
 */
public class Button {

    /**
     * Default constructor
     */
    public Button() {
    }

    /**
     * @return
     */
    public void pressRecordButton() {
        // TODO implement here
        return null;
    }

    /**
     * @return
     */
    public void pressReplayButton() {
        // TODO implement here
        return null;
    }

    /**
     * @return
     */
    public void pressDeleteButton() {
        // TODO implement here
        return null;
    }
}
```



```

/**
 * @return
 */
public void pressLockButton() {
    // TODO implement here
    return null;
}

/**
 * @return
 */
public void pressUnlockButton() {
    // TODO implement here
    return null;
}
}

```

```

import java.io.*;
import java.util.*;

/**
 *
 */
public class DigitalSoundRecorder {

    /**
     * Default constructor
     */
    public DigitalSoundRecorder() {
    }

    /**
     *
     */
    public Message[] messages;

    /**
     *
     */
    public int batteryLevel;

    /**
     *
     */
    public int maxMessageLength = 2;
}

```

```
/**
 *
 */
public int currentMessageSlot;

/**
 *
 */
public void Attribute1;

/**
 * @param slot
 * @param message
 * @return
 */
public void recordMessage(int slot, String message) {
    // TODO implement here
    return null;
}

/**
 * @param slot
 * @return
 */
public void replayMessage(int slot) {
    // TODO implement here
    return null;
}

/**
 * @param slot
 * @return
 */
public void deleteMessage(int slot) {
    // TODO implement here
    return null;
}

/**
 * @param slot
 * @return
 */
public void lockMessage(int slot) {
    // TODO implement here
    return null;
}

/**
 * @param slot
```

```

    * @return
    */
    public void unlockMessage(int slot) {
        // TODO implement here
        return null;
    }

    /**
     * @return
     */
    public boolean checkBattery() {
        // TODO implement here
        return false;
    }
}

```

```

import java.io.*;
import java.util.*;

/**
 *
 */
public class LCDDisplay {

    /**
     * Default constructor
     */
    public LCDDisplay() {
    }

    /**
     * @param level
     * @return
     */
    public void showBatteryStatus(int level) {
        // TODO implement here
        return null;
    }

    /**
     * @param message
     * @return
     */
    public void showMessage(String message) {
        // TODO implement here
        return null;
    }
}

```

```
}  
  
}
```

```
import java.io.*;  
import java.util.*;  
  
/**  
 *  
 */  
public class Message {  
  
    /**  
     * Default constructor  
     */  
    public Message() {  
    }  
  
    /**  
     *  
     */  
    public String messageText;  
  
    /**  
     *  
     */  
    public boolean isLocked;  
  
    /**  
     *  
     */  
    public int length;  
  
    /**  
     * @param text  
     * @param length  
     */  
    public Message(String text, int length) {  
        // TODO implement here  
    }  
  
    /**  
     * @return  
     */  
    public String getMessageText() {  
        // TODO implement here  
        return "";
```

```
}

/**
 * @param message
 * @return
 */
public void setMessageText(String message) {
    // TODO implement here
    return null;
}

/**
 * @return
 */
public int getLength() {
    // TODO implement here
    return 0;
}

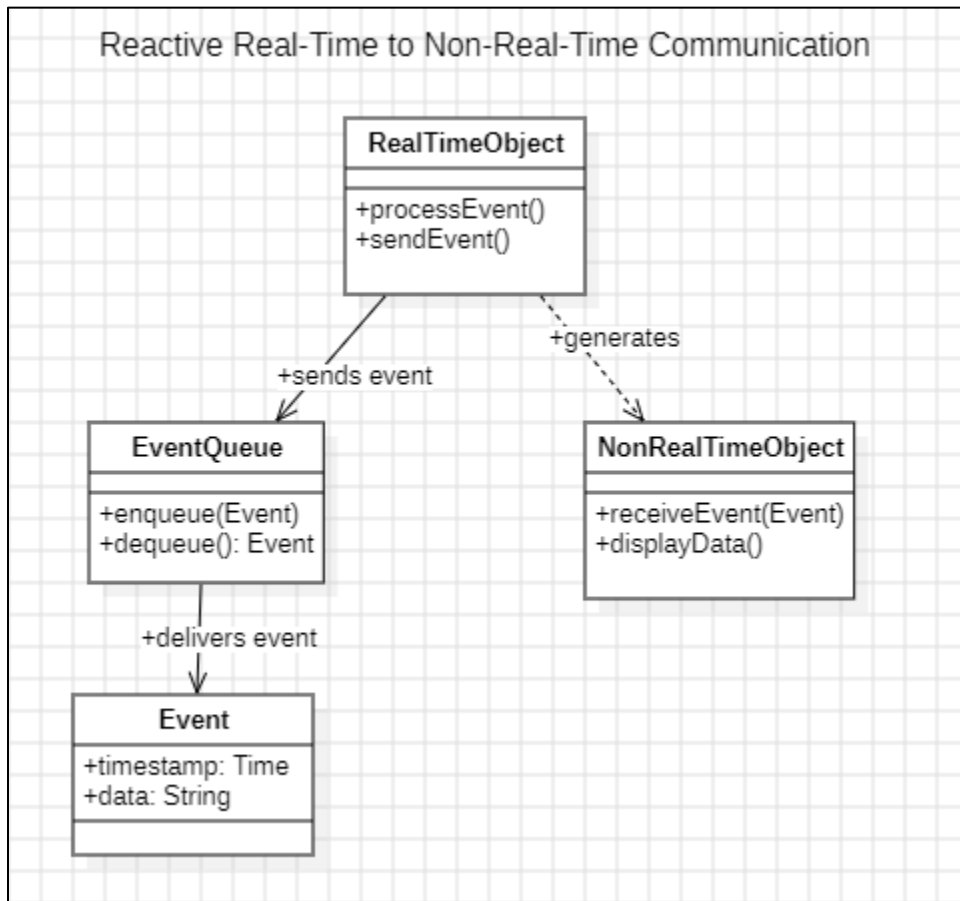
/**
 * @param length
 * @return
 */
public void setLength(int length) {
    // TODO implement here
    return null;
}

/**
 * @return
 */
public void lock() {
    // TODO implement here
    return null;
}

/**
 * @return
 */
public void unlock() {
    // TODO implement here
    return null;
}
}
```

AQ4 A reactive real-time object (e.g.: any embedded real time system) wants to communicate an event to an interactive non-real time object (e.g.: user-interface or system monitoring or data logging)?

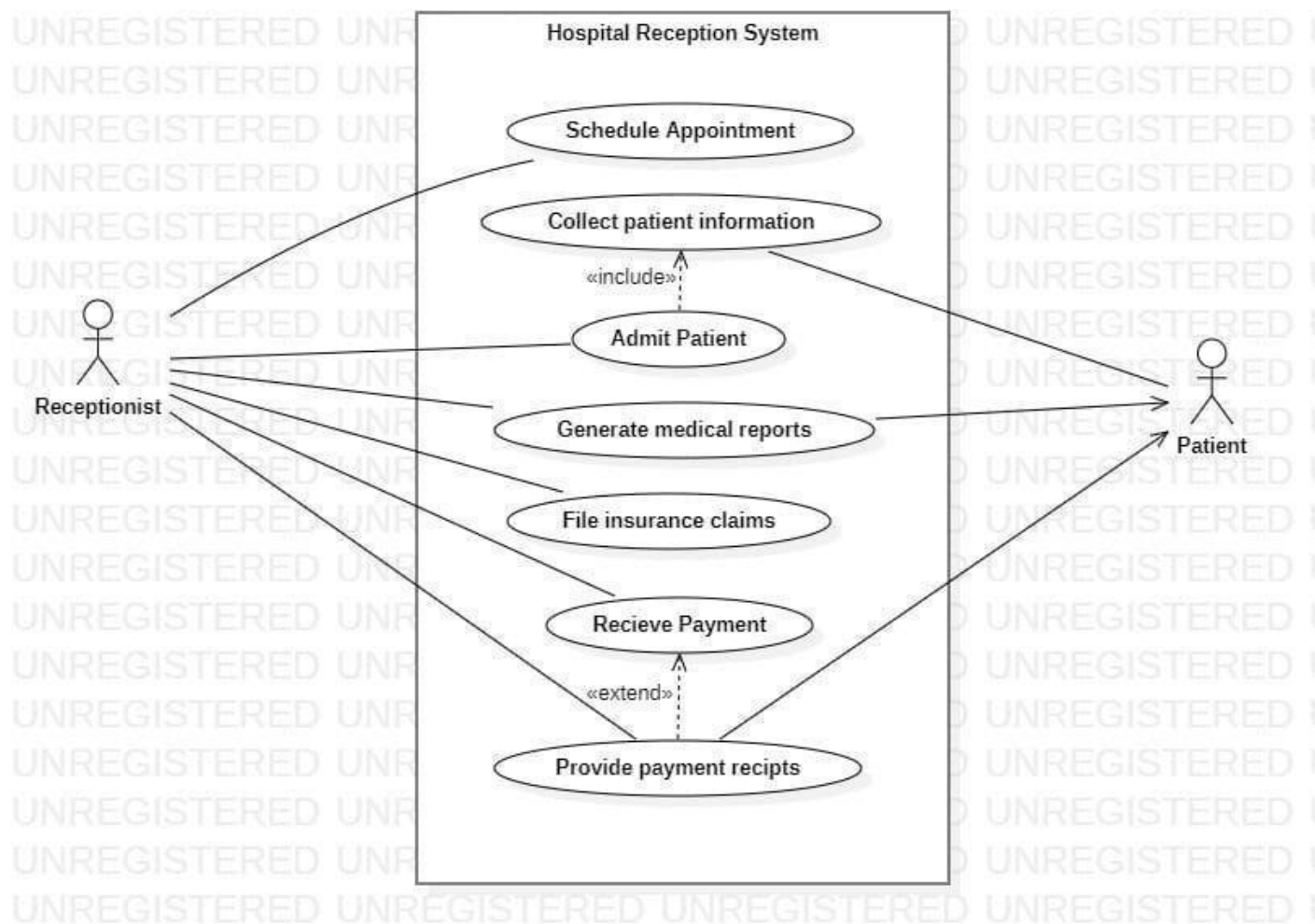
e.g., an embedded system sending sensor data to a monitoring UI



Q7 Purpose: Describe major services (functionality) provided by a hospital's reception.

Summary: Hospital Management System is a large system including several subsystems or modules providing variety of functions. Hospital Reception subsystem or module supports some of the many job duties of hospital receptionist. Receptionist schedules patient's appointments and admission to the hospital, collects information from patient upon patient's arrival and/or by phone. For the patient that will stay in the hospital ("inpatient") she or he should have a bed allotted in a ward. Receptionist may also receive and process patient's payments, record them and provide receipts. She also is responsible to file insurance claims and generate medical reports.

a. Make the use case diagram

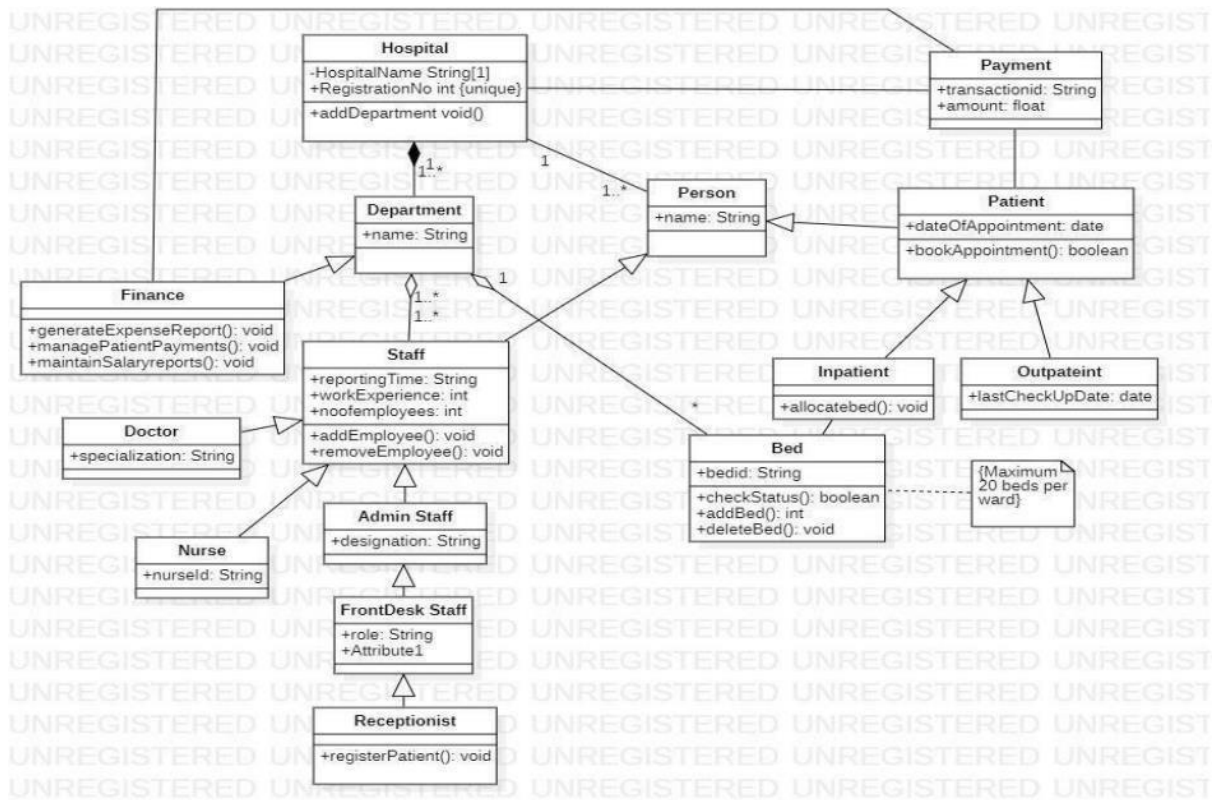


b. Provide description for at least 04 use cases

USE CASES	DESCRIPTION
Schedule Appointment	The Schedule Appointment use case involves the Receptionist and Patient. The process begins when the patient provides necessary details such as name, contact information, and reason for the visit. The receptionist checks the doctor's availability

	and selects an appropriate time slot. Once confirmed, the appointment is recorded in the hospital's system, and a confirmation is sent to the patient. The precondition for this use case is that the patient must provide the required information. The postcondition ensures that the appointment is successfully stored in the hospital database.
Admit Patient	The Admit Patient use case is managed by the Receptionist when a patient requires hospitalization. The process starts by collecting the patient's personal and medical details. The receptionist verifies room availability and assigns a ward and bed to the patient. Once the admission details are recorded in the system, the patient is officially admitted. The precondition for this use case is that the patient must require admission and a room must be available. The postcondition ensures that the patient is admitted and assigned a bed in the hospital.
Process Payment	The Process Payment use case involves both the Receptionist and the Patient. The process starts when the receptionist checks the patient's outstanding balance. The patient then makes the payment using cash, card, or insurance. The receptionist records the payment transaction in the system and generates a receipt for the patient. The precondition for this use case is that the patient must have a pending payment. The postcondition ensures that the payment is successfully processed and updated in the system.
File Insurance Claim	The File Insurance Claim use case involves the Receptionist and the InsuranceCompany. If a patient has valid insurance coverage, the receptionist verifies the details and generates an insurance claim form. The form is then submitted to the insurance company for processing. Once the claim is filed, the receptionist updates the claim status in the hospital's system. The precondition for this use case is that the patient must have valid insurance coverage. The postcondition ensures that the insurance claim is successfully submitted and recorded in the hospital database.

c. Make the class diagram



d. Generate Java stub code

java file generator > Model > Model				
Name	Date modified	Type	Size	
Appointment	05-02-2025 16:29	Java Source File	1 KB	
InsuranceCompany	05-02-2025 16:29	Java Source File	1 KB	
MedicalReport	05-02-2025 16:29	Java Source File	1 KB	
Patient	05-02-2025 16:29	Java Source File	1 KB	
Payment	05-02-2025 16:29	Java Source File	1 KB	
Receptionist	05-02-2025 16:29	Java Source File	2 KB	
Ward	05-02-2025 16:29	Java Source File	1 KB	

```

import java.io.*;
import java.util.*;

/**
 *
 */
public class Appointment {

    /**
     * Default constructor
     */
    public Appointment() {
    }
  
```

```

/**
 *
 */
public int {unique} appointmentID;

/**
 *
 */
public Date date;

/**
 *
 */
public Time time;

/**
 *
 */
public String doctorName;

/**
 * @return
 */
public boolean scheduleAppointment() {
    // TODO implement here
    return false;
}
}

```

```

import java.io.*;
import java.util.*;

/**
 *
 */
public class InsuranceCompany {

    /**
     * Default constructor
     */
    public InsuranceCompany() {
    }

    /**
     *
     */

```

```
public int {unique} companyID;

/**
 *
 */
public String name;

/**
 * @param p
 * @return
 */
public boolean processClaim(Patient p) {
    // TODO implement here
    return false;
}
}
```

```
import java.io.*;
import java.util.*;

/**
 *
 */
public class MedicalReport {

    /**
     * Default constructor
     */
    public MedicalReport() {
    }

    /**
     *
     */
    public int {unique} reportID;

    /**
     *
     */
    public String diagnosis;

    /**
     * @return
     */
    public void generateReport() {
        // TODO implement here
    }
}
```

```
    return null;
}
}
```

```
import java.io.*;
import java.util.*;

/**
 *
 */
public class Patient {

    /**
     * Default constructor
     */
    public Patient() {
    }

    /**
     *
     */
    public int {unique} patientID;

    /**
     *
     */
    public String name;

    /**
     *
     */
    public String contact;

    /**
     *
     */
    public String address;

    /**
     *
     */
    public boolean admitted;

    /**
     * @param amount
     * @return
     */
}
```

```
    */  
    public void makePayment(double amount) {  
        // TODO implement here  
        return null;  
    }  
}
```

```
import java.io.*;  
import java.util.*;  
  
/**  
 *  
 */  
public class Payment {  
  
    /**  
     * Default constructor  
     */  
    public Payment() {  
    }  
  
    /**  
     *  
     */  
    public int {unique} paymentID;  
  
    /**  
     *  
     */  
    public double amount;  
  
    /**  
     *  
     */  
    public String paymentType;  
  
    /**  
     * @return  
     */  
    public boolean processPayment() {  
        // TODO implement here  
        return false;  
    }  
}
```

```
import java.io.*;
import java.util.*;

/**
 *
 */
public class Receptionist {

    /**
     * Default constructor
     */
    public Receptionist() {
    }

    /**
     *
     */
    public int {unique} receptionistID;

    /**
     *
     */
    public String name;

    /**
     * @param p
     * @param a
     * @return
     */
    public boolean scheduleAppointment(Patient p, Appointment a) {
        // TODO implement here
        return false;
    }

    /**
     * @param p
     * @param w
     * @return
     */
    public boolean admitPatient(Patient p, Ward w) {
        // TODO implement here
        return false;
    }

    /**
     * @param p
     * @param amount
```

```

    * @return
    */
    public void processPayment(Patient p, double amount) {
        // TODO implement here
        return null;
    }

    /**
     * @param p
     * @param ic
     * @return
     */
    public boolean fileInsuranceClaim(Patient p, InsuranceCompany ic) {
        // TODO implement here
        return false;
    }
}

```

```

import java.io.*;
import java.util.*;

/**
 *
 */
public class Ward {

    /**
     * Default constructor
     */
    public Ward() {
    }

    /**
     *
     */
    public int {unique} wardID;

    /**
     *
     */
    public String roomNumber;

    /**
     *
     */
    public int {>0} bedCount;

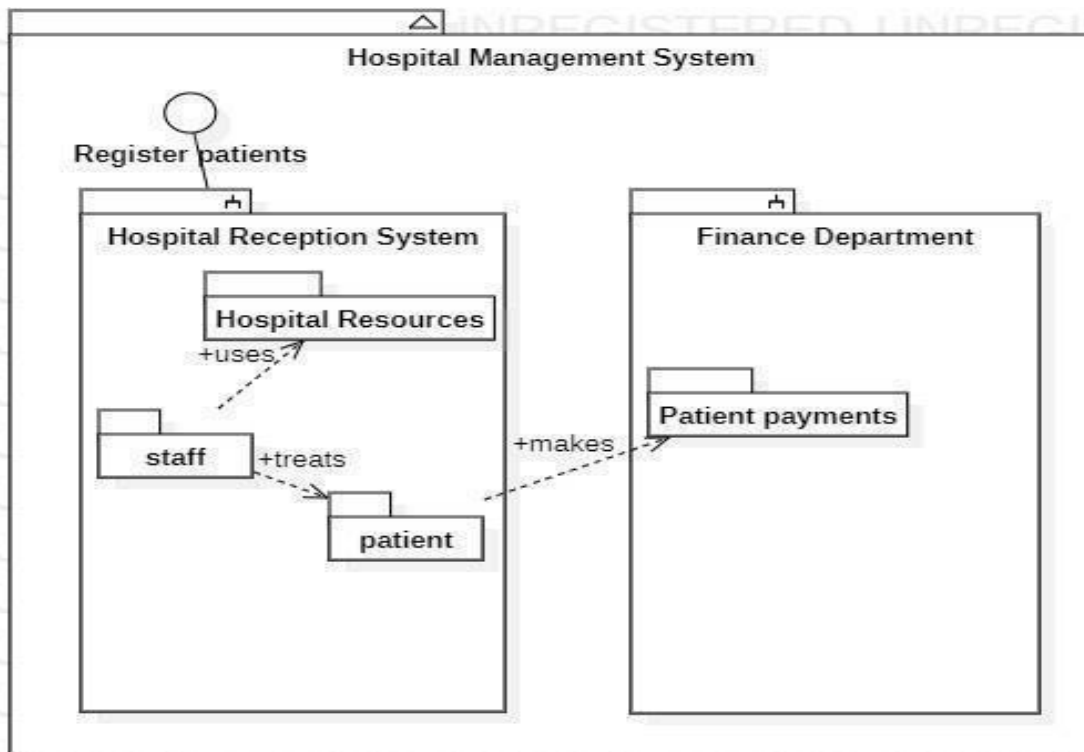
```

```

/**
 *
 */
public boolean isAvailable;
}

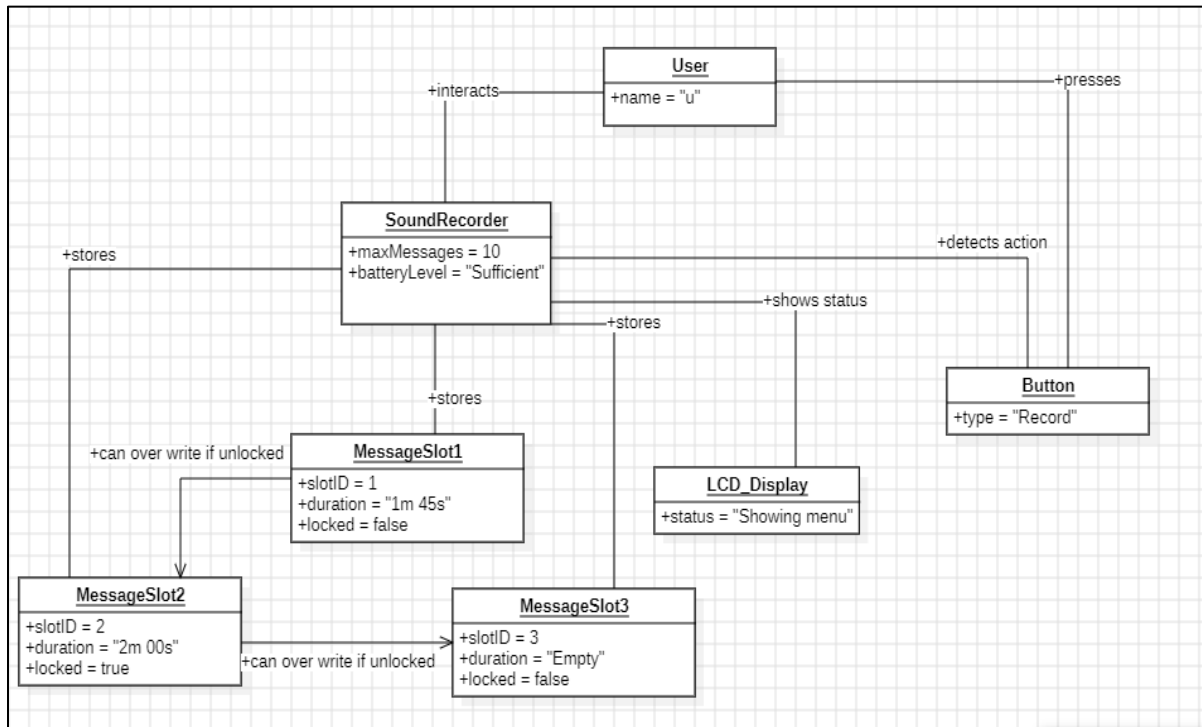
```

e. Create Object Diagram

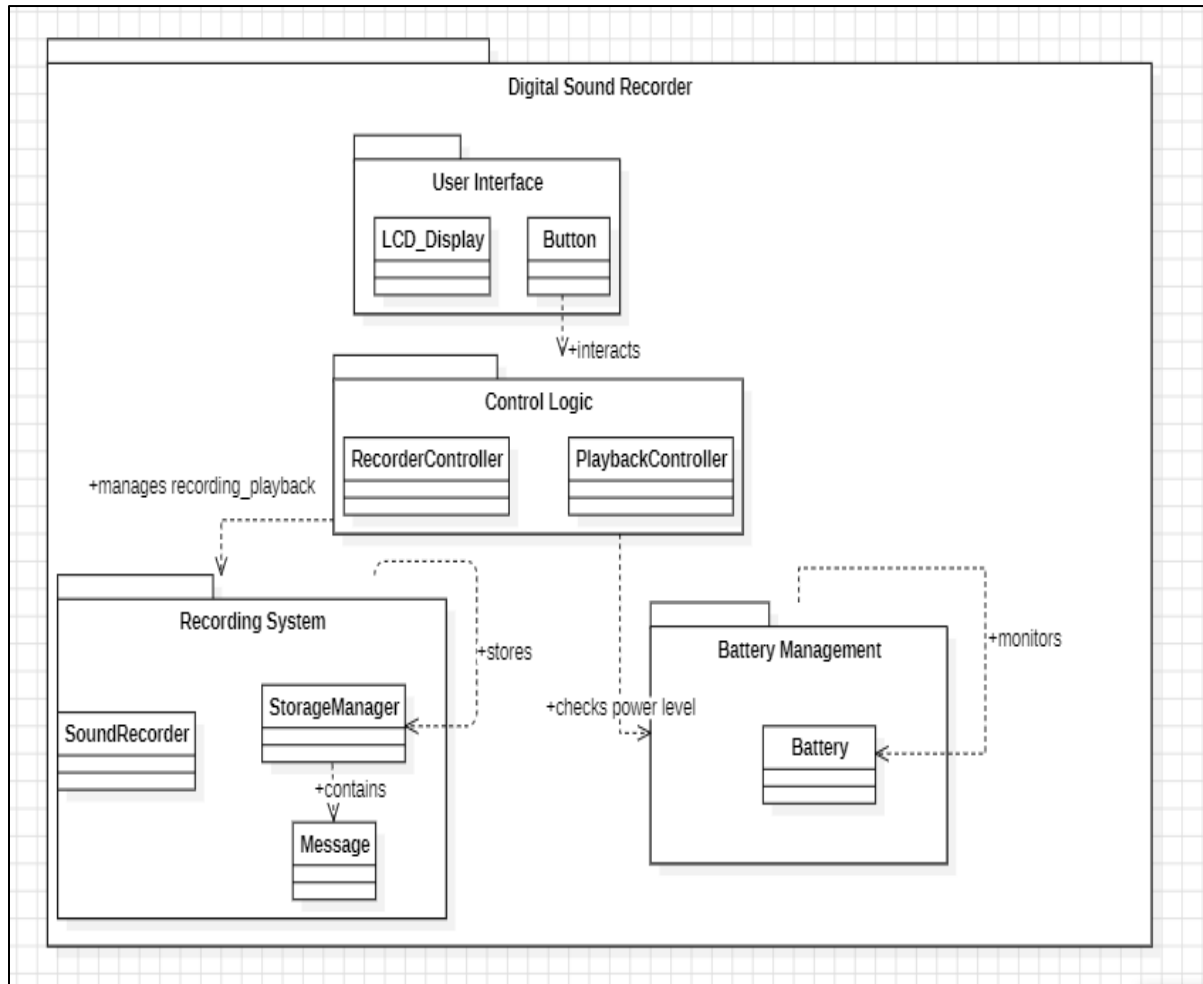


Q8 Create an object diagram for a Digital Sound Recorder with the following main features:

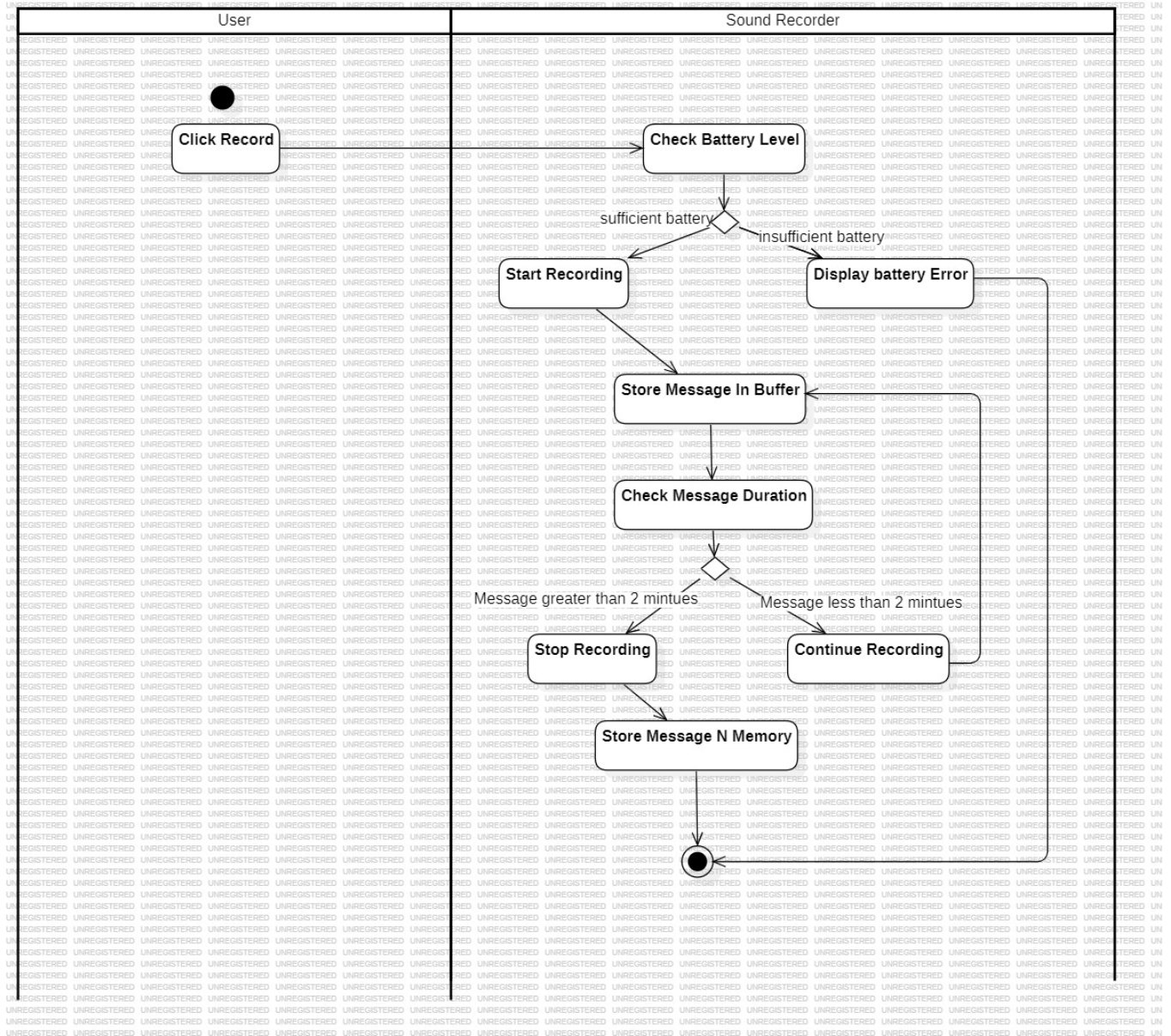
- 1. The recorder stores up to 10 messages**
 - 2. Each message is max. 2 minutes long**
 - 3. The user can record message**
 - 4. Recording of a message end after 2 minutes or when the user stops recording**
 - 5. Recording destroys the original message at chosen slot**
 - 6. Sufficient level of battery is checked before recording message**
 - 7. Message of a given slot can be replayed**
 - 8. Sufficient level of battery is checked before replaying message**
 - 9. Messages can be locked/unlocked**
 - 10. Locked messages cannot be deleted or over-written by recording to the same slot**
- User uses LCD display and buttons to interact with recorder**



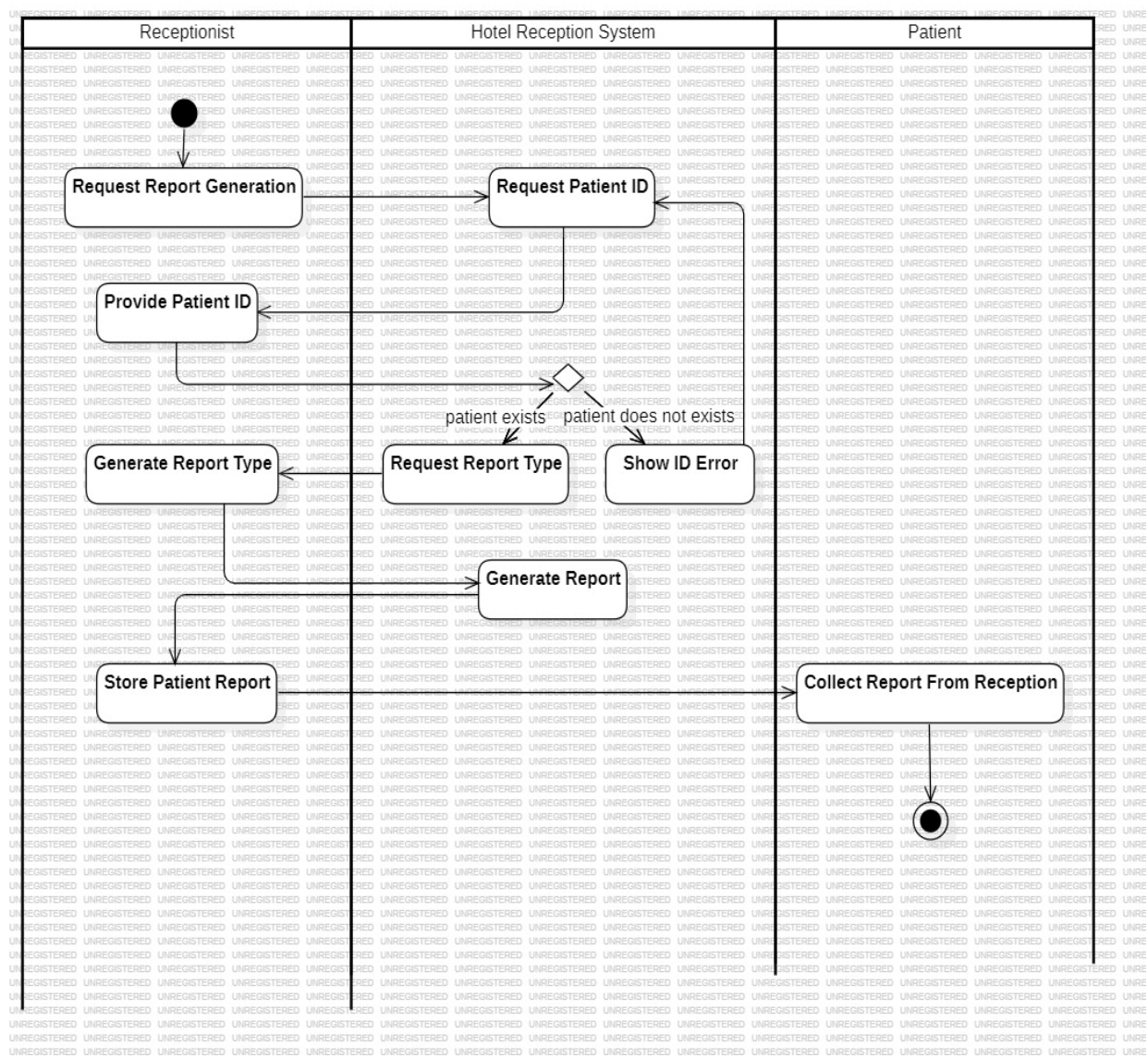
Q9 Create a package diagram for the Digital Sound Recorder in AP1



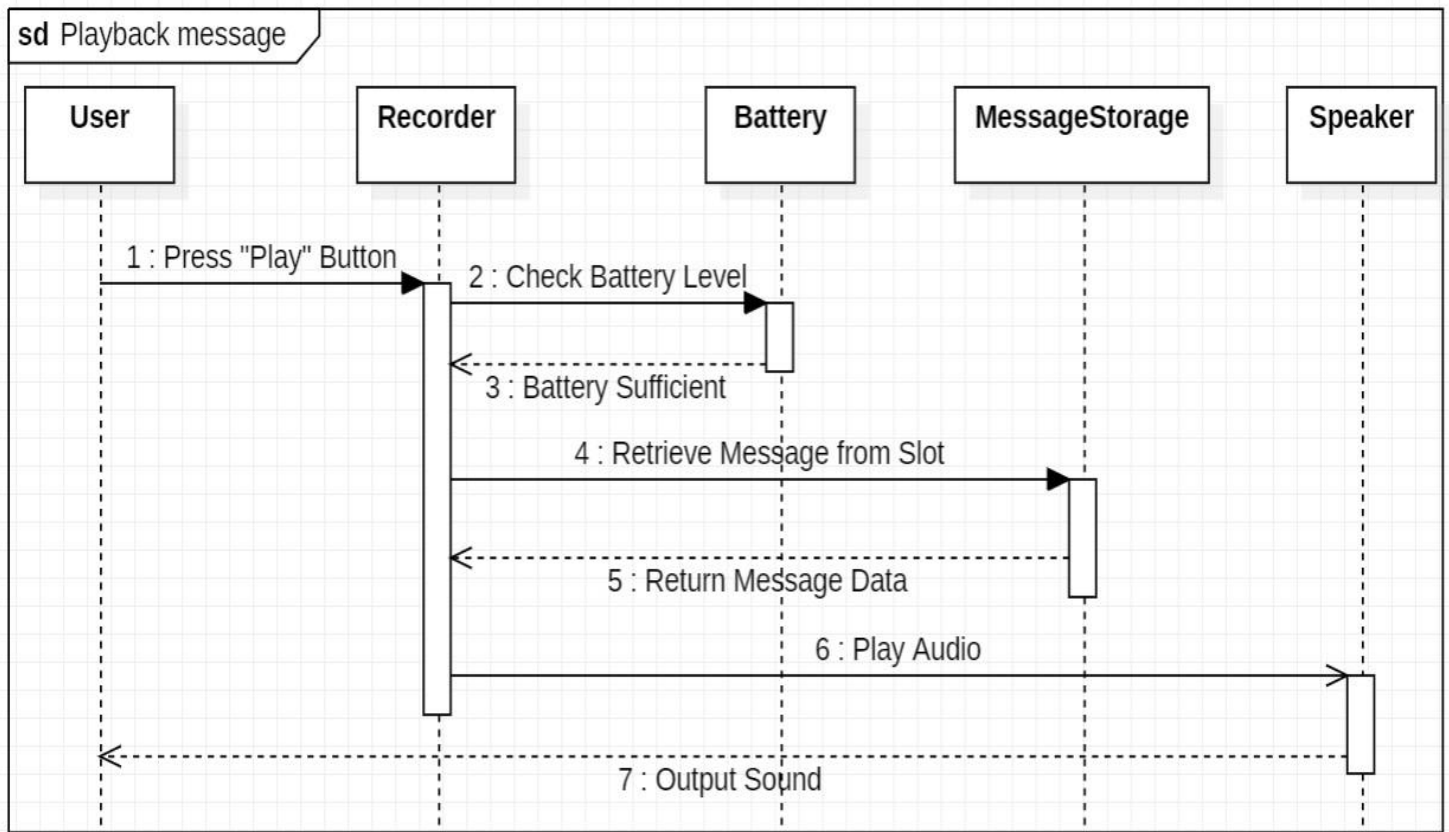
Q10 Create an activity diagram for record message use case of AP1?



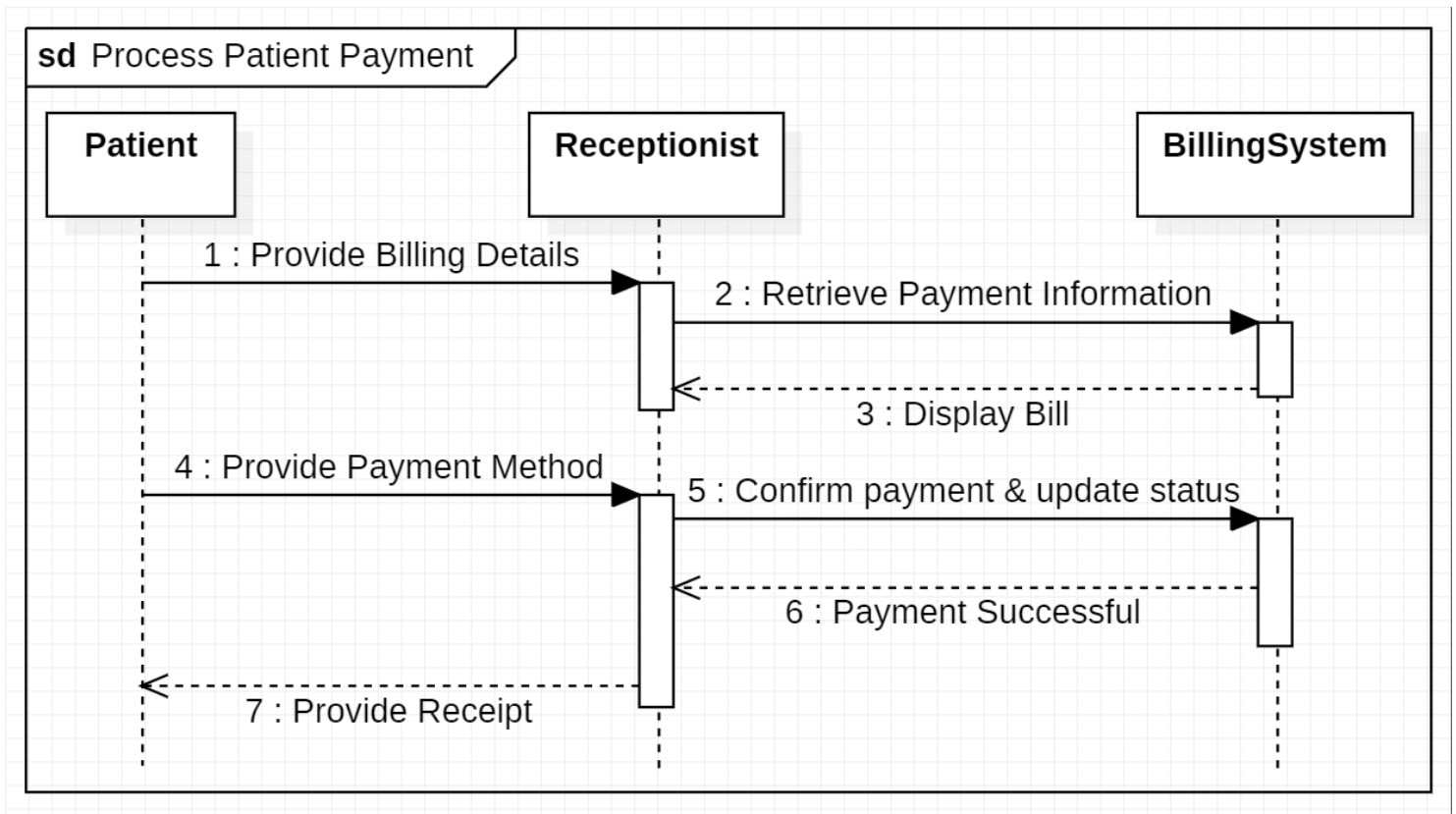
Q11 Create an activity diagram for any two use cases of Hospital Reception Subsystem in CP1



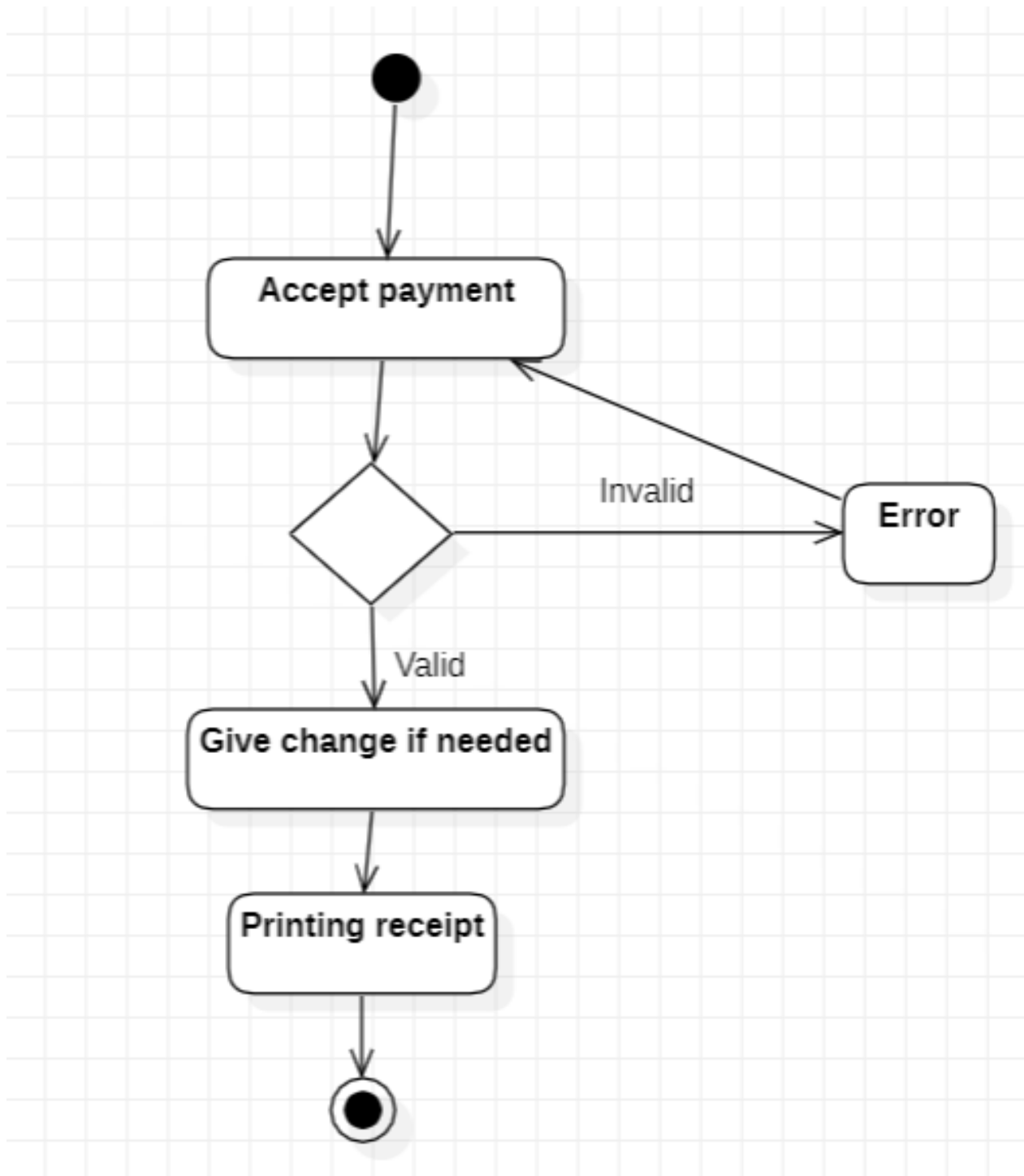
Ques 12: Create a sequence diagram for the playback message use case of AP1?



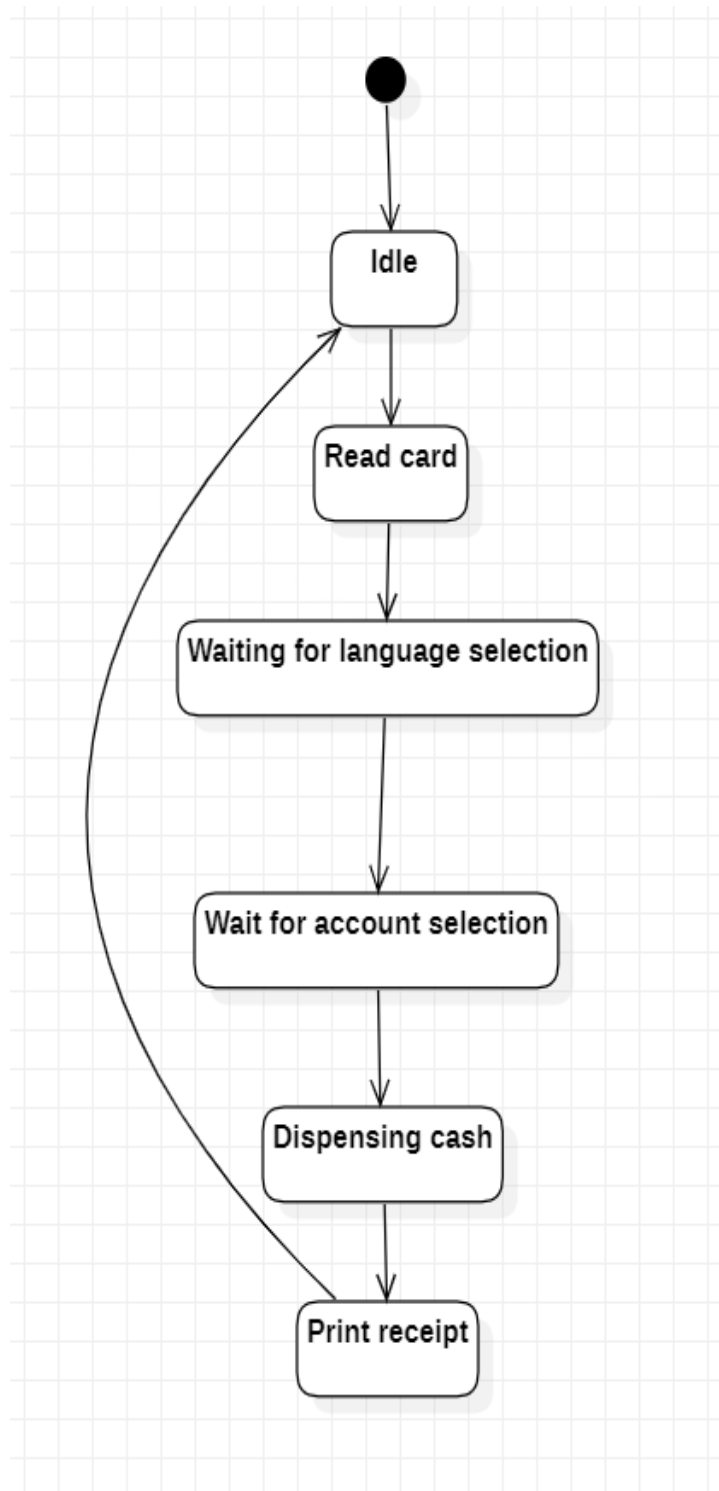
Q 13 Create a sequence diagram for any two use cases of Hospital Reception Subsystem in CP1



Q14 Create a state diagram for any two use cases of Hospital Reception Subsystem in CP1

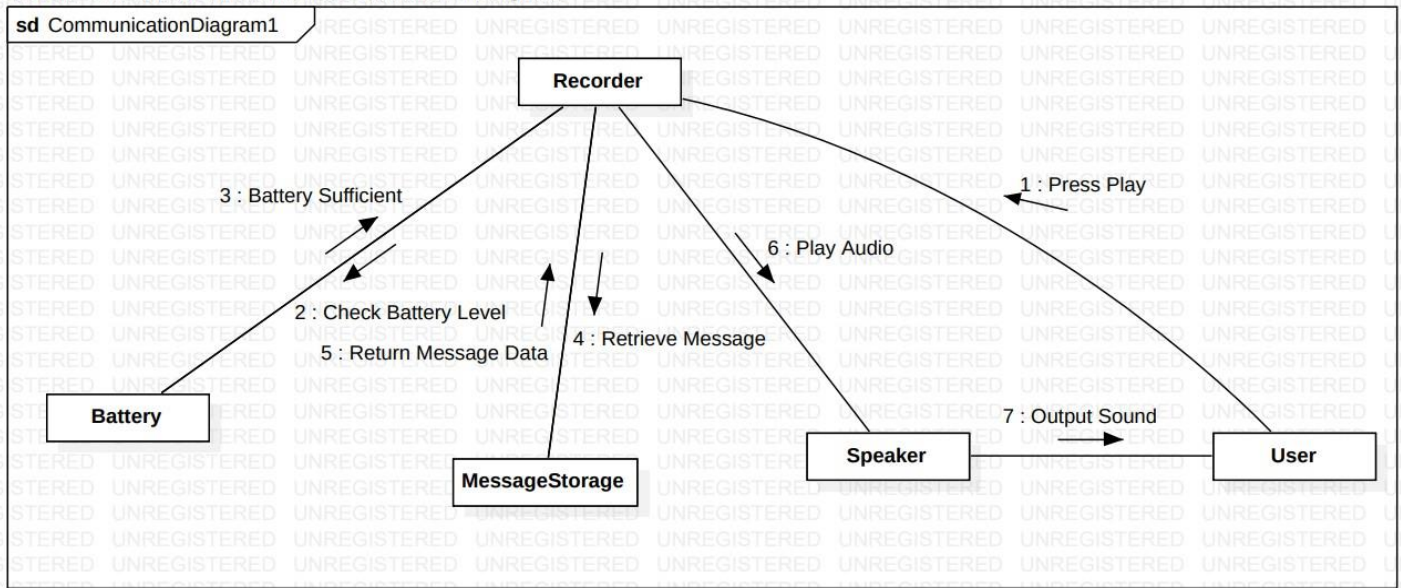


Q 15 Create a state diagram for a BANK ATM machine?

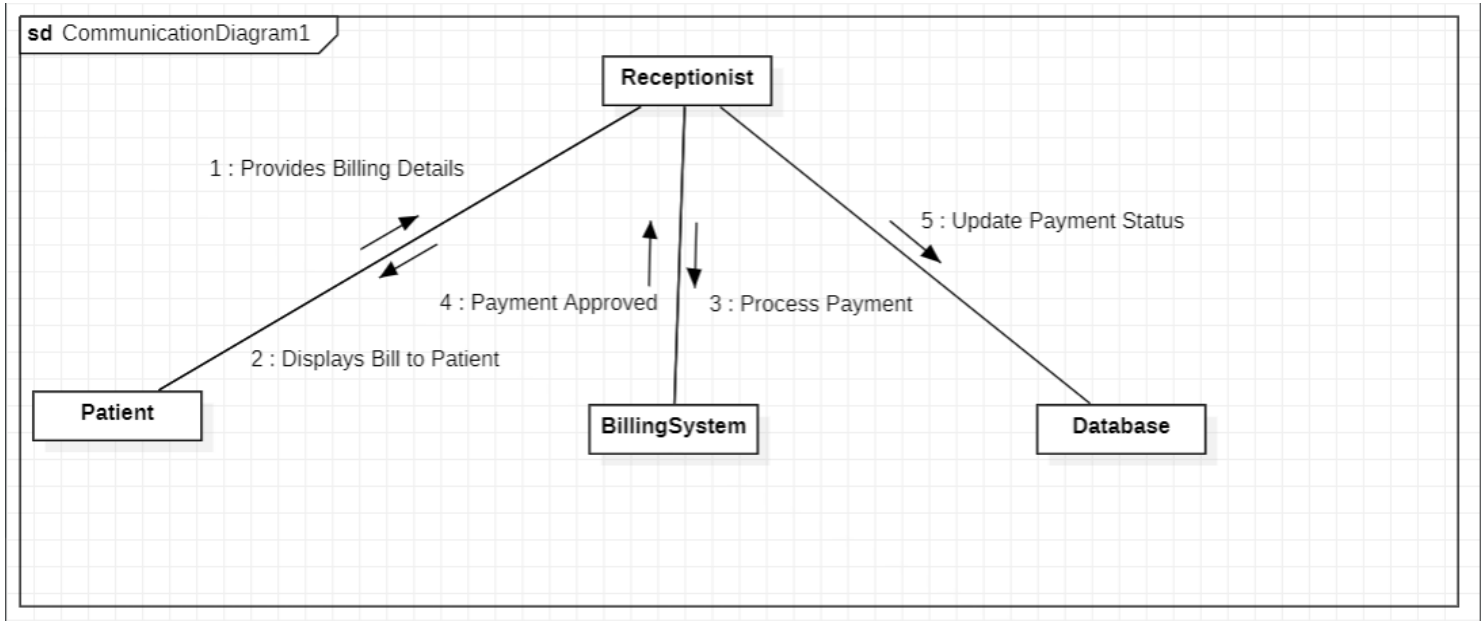


Q 16 Create a communication diagram for the playback message use case of API?

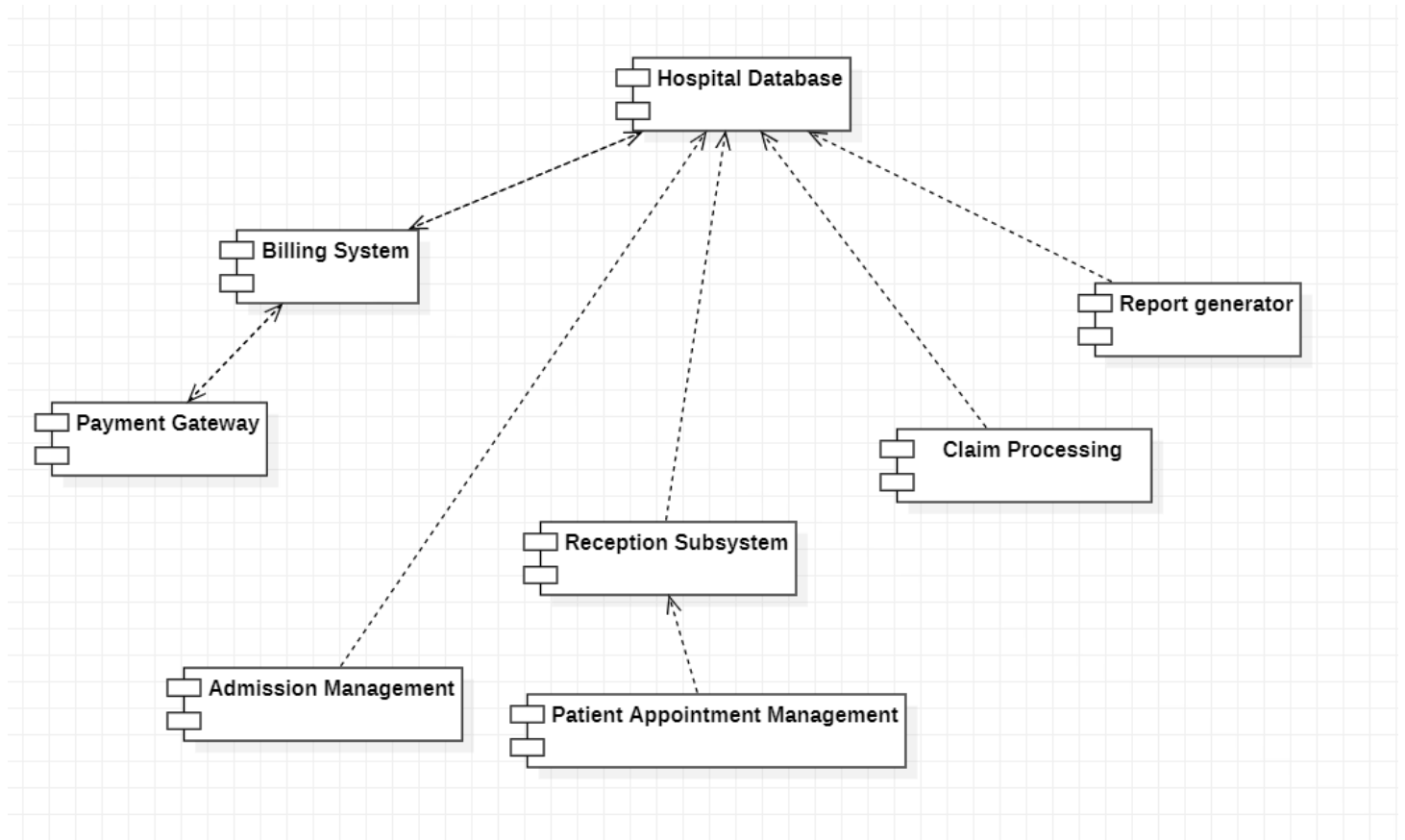
Model::Collaboration1::Interaction1::CommunicationDiagram1



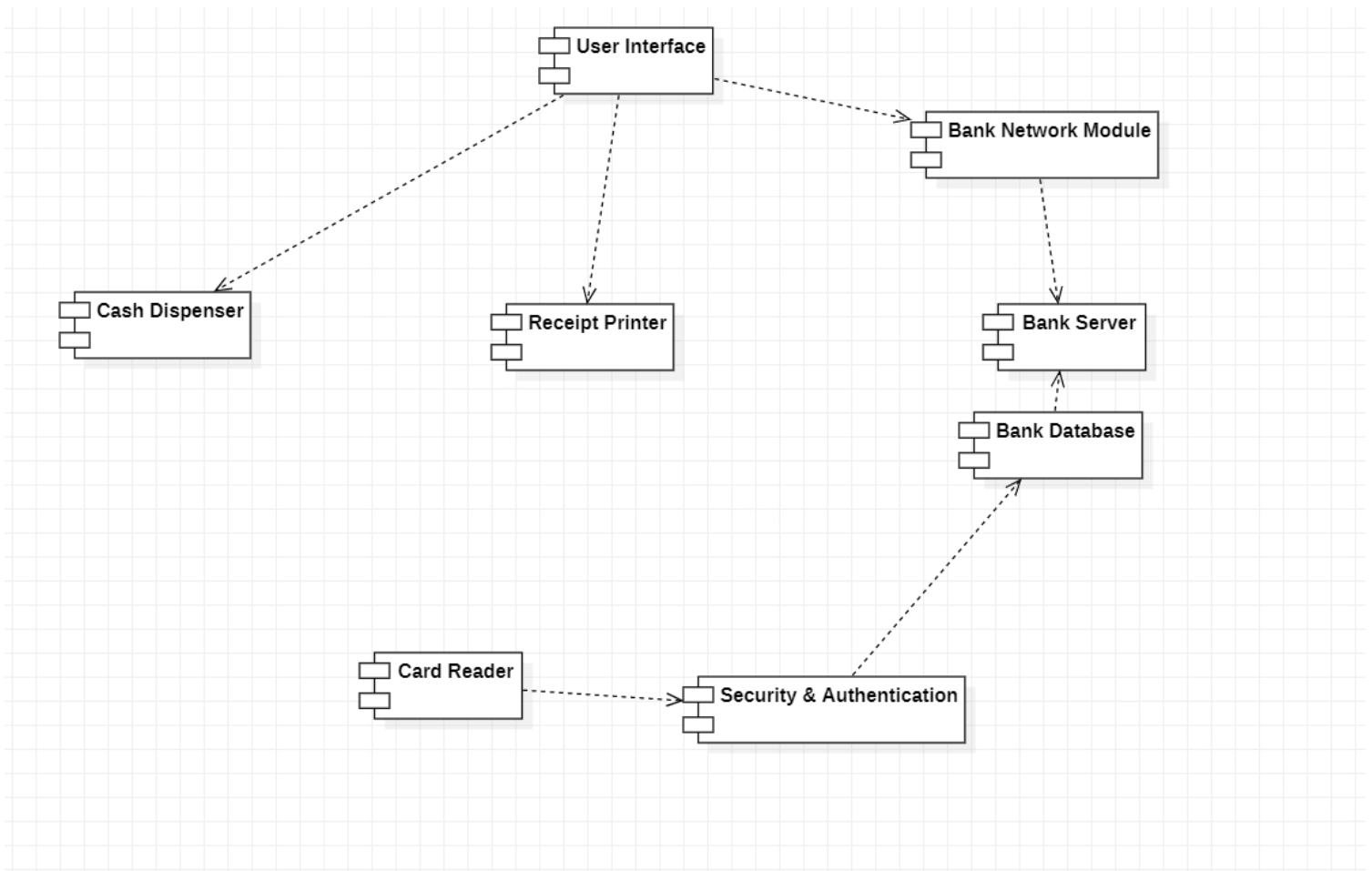
Q 17 Create a communication diagram for any two use cases of Hospital Reception Subsystem in CP1?



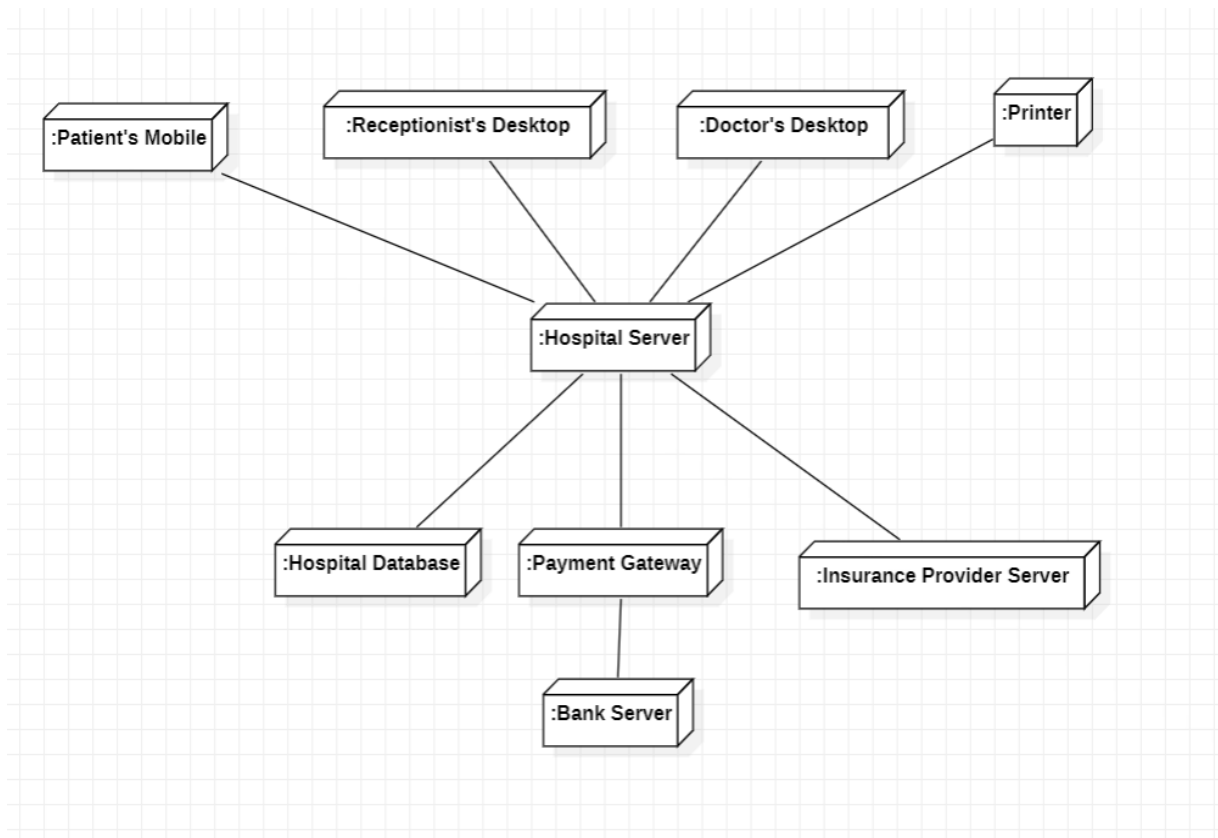
Q 18 Create a component diagram for the Hospital Reception Subsystem?



Q 19 Create a component diagram for ATM Machine.



Q 20 Create a deployment diagram for the Hospital Reception Subsystem.



Q 21 Create a deployment diagram for ATM machine.

